

S22 Comprehensive Statistical Inference Specification

1. Epistemic Framework

This study follows a specification-internal evaluation paradigm, in which all labels are generated directly from the governance specification rather than from external real-world ground truth. Consequently, all reported statistics are descriptive measures of boundary recoverability under controlled synthetic perturbation conditions.

No inferential claims regarding real-world operational populations are intended.

2. Hierarchical Bootstrap Procedure

For each reported metric (AUROC, CDI, latency, memory utilisation, and related measures), a hierarchical bootstrap procedure was implemented.

Level 1 – Seed-Level Resampling

For each of the 10 random seeds, 10,000 stratified bootstrap resamples of the 48-sample synthetic test set were generated with replacement.

Level 2 – Seed Pooling

The resulting seed-level bootstrap distributions were combined using Rubin's Rules (Rubin, 1987):

- Point estimate = mean of seed medians
- Variance = within-seed variance + between-seed variance

Interval Reporting

The 2.5th and 97.5th percentiles of the combined bootstrap distribution are reported as 95% bootstrap confidence bands rather than classical frequentist confidence intervals.

No p-values were computed.

3. Binomial Confidence Intervals

For governance-gate satisfaction proportions, exact binomial (Clopper–Pearson) 95% confidence intervals were calculated.

4. Performance Metrics

AUROC

Computed using:

```
roc_auc_score(multi_class='ovr')
```

from scikit-learn.

Macro-Averaged F1

Computed using:

```
f1_score(average='macro')
```

Sensitivity and Specificity

Computed at:

- CDI threshold = 0.70
- Youden-optimised thresholds where applicable

Expected Calibration Error (ECE)

ECE was computed using 10 equal-width probability bins across the interval [0,1].

5. Correlation Structure and Noise Model

Inter-feature dependencies were induced using a Gaussian copula with correlation matrix ρ derived from publicly available incident statistics.

Additive Gaussian noise was injected after feature normalisation with: $\sigma = 0.10$

Sensitivity analyses for: $\sigma \in \{0.05, 0.10, 0.15, 0.20\}$ are reported in Section 5.3 of the main manuscript.

6. Software Environment

Software Package	Version
Python	3.11.8
scikit-learn	1.3.0
NumPy	1.26.4
SciPy	1.12.0
PyTorch	2.1.0

7. Interpretation Conventions

All reported confidence intervals should be interpreted as descriptive bootstrap variability bands rather than probabilistic coverage intervals for unknown population parameters.

Accordingly, the manuscript intentionally avoids inferential terminology such as:

- “statistically significant”
- “evidence for”
- “proof of”

and instead adopts descriptive reporting language.

Example:

“The calibrated model recovered the governance boundary with AUROC = 0.75.”

8. Data and Code Availability

The synthetic data generator, bootstrap procedures, and statistical analysis notebooks are archived in GitHub: <https://github.com/oluseun-akeredolu/HIMS-CDI>